

ELE456
Lecture Notes

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About these notes

These notes develop the mathematical foundations of robotics, from linear algebra to models of robot motion, sensing, and planning. Each chapter introduces a topic through definitions, explanations, and worked examples. The material builds progressively, with an emphasis on understanding the assumptions behind each calculation and interpreting the resulting relationships.

Preparation. These notes are prepared with AI assistance from Paolo Stegagno's lecture transcripts and board material and reviewed by the instructor. Errors may remain; please report suspected mistakes.

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Contents

1	Introduction and linear algebra	1
1.1	Matrices and dimensions	1
1.2	Matrix addition	1
1.3	Matrix multiplication	2
1.4	The transpose	3
1.5	Identity matrices and inverses	3
1.6	Orthogonal matrices	4
1.7	Key identities	5
1.8	Additional exercises	6
2	Reference frames and planar rotations	7
2.1	Coordinates depend on a reference frame	7
2.2	Constructing the rotation matrix	8
2.3	Transforming the coordinates of a point	9
2.4	Reversing a coordinate transformation	11
2.5	Composing rotations through several frames	11
2.6	The limit of a rotation-only model	12
2.7	Key identities	13
2.8	Additional exercises	14
3	Planar rigid transformations and homogeneous coordinates	15
3.1	Rotation and translation between two frames	15
3.2	Reversing the affine relationship	17
3.3	Composing transformations through a sensor frame	18
3.4	Homogeneous coordinates and transformation matrices	18
3.5	Composition by matrix multiplication	19

3.6	Inverting a homogeneous transformation	20
3.7	Key identities	21
3.8	Additional exercises	22
4	Three-dimensional rotations and orientation representations	23
4.1	Spatial frames and rotation matrices	23
4.2	Elementary rotations	24
4.3	What makes a matrix a valid spatial rotation?	25
4.4	Roll, pitch, and yaw as a sequence of rotations	26
4.5	Worked example	27
4.6	Recovering roll, pitch, and yaw	28
4.7	Choosing and maintaining an orientation representation	29
4.8	Key identities	30
4.9	Additional exercises	31
5	Three-dimensional rigid transformations and pose	32
5.1	Combining spatial rotation and translation	32
5.2	The reverse coordinate transformation	33
5.3	Homogeneous transformations in space	34
5.4	Inverting a homogeneous transformation	34
5.5	What rounding changes	35
5.6	Position, orientation, and pose	36
5.7	Additional exercises	37
6	Robot systems and sensing	38
6.1	Sensing, processing, and action	38
6.2	Sensors and the environment	38
6.3	Actuators and physical interaction	39
6.4	Connecting measurements to commands	39
A	Rotation representation conversions	41

Chapter 1

Introduction and linear algebra

Robotics brings together mathematical models of motion, measurements from sensors, and algorithms that determine how a robot should act. Describing a robot's position, relating measurements to its surroundings, and planning its motion all require a precise mathematical language. Linear algebra provides many of the tools used to express these relationships.

This chapter introduces matrices and the operations needed to work with them: addition, multiplication, transposition, and inversion. It then examines orthogonal matrices, whose inverses can be obtained simply by transposing. The examples develop two habits that will remain essential throughout the study of robotics: checking dimensions before performing an operation and preserving the order of factors in a matrix product.

1.1 Matrices and dimensions

A *matrix* is an array of numbers arranged in rows and columns. Its dimensions are written as
number of rows $\times$ number of columns.

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}. \quad (1.1)$$

It has three rows and two columns. In general, a_{ij} denotes the entry in row i , column j ; for example, $a_{21} = 3$ and $a_{32} = 1$.

1.2 Matrix addition

Two matrices can be added only if they have the same dimensions. Their entries are then added in corresponding positions:

$$(A + C)_{ij} = a_{ij} + c_{ij}.$$

Consider two additional matrices,

$$B = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

The sum $A + B$ is undefined because the dimensions differ. The sum $A + C$ is defined, and its calculation is

$$A + C = \begin{bmatrix} 1+0 & 2+1 \\ 3+0 & 1+0 \\ 0+1 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \\ 1 & 1 \end{bmatrix}. \quad (1.2)$$

Addition is commutative when defined: $A + C = C + A$.

1.3 Matrix multiplication

1.3.1 Dimension rule and row-by-column calculation

For matrices $M \in \mathbb{R}^{m \times n}$ and $N \in \mathbb{R}^{n \times p}$, the product MN is defined and has dimensions $m \times p$:

$$\underbrace{(m \times n)(n \times p)}_{\text{inner dimensions must match}} \longrightarrow m \times p.$$

The entry in row i , column j is the sum of the products of corresponding entries of row i of M and column j of N :

$$(MN)_{ij} = \sum_{k=1}^n m_{ik} n_{kj}. \quad (1.3)$$

Matrix multiplication is therefore a row-by-column operation, rather than entrywise multiplication.

For the matrices A and B above, the dimensions are $(3 \times 2)(2 \times 2)$, so AB is a 3×2 matrix:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad (1.4)$$

$$= \begin{bmatrix} 1 \cdot 1 + 2 \cdot 1 & 1 \cdot 2 + 2 \cdot 1 \\ 3 \cdot 1 + 1 \cdot 1 & 3 \cdot 2 + 1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 & 0 \cdot 2 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & 7 \\ 1 & 1 \end{bmatrix}. \quad (1.5)$$

1.3.2 The order of the factors matters

Reversing the factors gives the dimensions $(2 \times 2)(3 \times 2)$. The inner dimensions do not match, so BA is undefined, even though AB exists.

Even when both orders are defined, the results need not agree. With

$$E = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

direct multiplication gives

$$BE = \begin{bmatrix} 7 & 10 \\ 4 & 6 \end{bmatrix}, \quad EB = \begin{bmatrix} 3 & 4 \\ 7 & 10 \end{bmatrix}.$$

Thus $BE \neq EB$. Matrix multiplication is not commutative in general. Some particular matrices do commute, but factors cannot be exchanged without a reason that applies to the matrices in question.

1.4 The transpose

Transposing a matrix interchanges its rows and columns. In entry notation, $(M^T)_{ij} = m_{ji}$. If M has dimensions $m \times n$, then M^T has dimensions $n \times m$. For the running example,

$$A^T = \begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 3}. \quad (1.6)$$

The transpose of a product reverses the order of its factors:

$$\boxed{(AB)^T = B^T A^T.} \quad (1.7)$$

The dimensions provide a quick check. Here $B^T A^T$ has dimensions $(2 \times 2)(2 \times 3)$, producing a 2×3 matrix as required. The incorrect expression $A^T B^T$ would have dimensions $(2 \times 3)(2 \times 2)$ and is undefined.

Using the matrices defined above, this identity can be verified directly:

$$B^T A^T = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 1 \\ 4 & 7 & 1 \end{bmatrix} = (AB)^T.$$

Dimension checks can rule out an invalid expression, but matching dimensions alone do not prove that an identity is correct.

1.5 Identity matrices and inverses

The identity matrix I_n has ones on its main diagonal and zeros elsewhere. For a square matrix $M \in \mathbb{R}^{n \times n}$, an inverse M^{-1} satisfies

$$MM^{-1} = M^{-1}M = I_n. \quad (1.8)$$

This ordinary, two-sided inverse is defined only for square matrices, and not every square matrix has one. The condition for invertibility is

$$\boxed{\det(M) \neq 0.} \quad (1.9)$$

A square matrix with zero determinant is called *singular*.

1.5.1 A one-by-one example

For $H = [5]$, the inverse is $H^{-1} = [1/5]$, since

$$HH^{-1} = H^{-1}H = [1] = I_1.$$

This is the familiar reciprocal operation for a nonzero scalar.

1.5.2 A diagonal example and its determinant

The second example is

$$G = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad G^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}.$$

Multiplying verifies the inverse:

$$GG^{-1} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot \frac{1}{2} \\ 0 \cdot 1 + 2 \cdot 0 & 0 \cdot 0 + 2 \cdot \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

The reverse product also equals I_2 . For a general 2×2 matrix,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Hence $\det(G) = 1 \cdot 2 - 0 \cdot 0 = 2 \neq 0$, as required. Reciprocating the diagonal entries works here because G is diagonal with nonzero diagonal entries; matrix inversion is not generally an entrywise reciprocal operation.

1.6 Orthogonal matrices

A real square matrix L is *orthogonal* if its inverse equals its transpose:

$$\boxed{L^{-1} = L^{\mathsf{T}}.} \tag{1.10}$$

Equivalently, $LL^{\mathsf{T}} = L^{\mathsf{T}}L = I$. This gives a practical way to check orthogonality: form the transpose, multiply, and verify the identity matrix. For an orthogonal matrix, an inverse can be obtained by transposing instead of performing a general inverse calculation.

1.6.1 Verifying orthogonality

Consider

$$L = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}, \quad L^{\mathsf{T}} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}.$$

Their product is

$$LL^{\mathsf{T}} = \begin{bmatrix} \frac{1}{4} + \frac{3}{4} & -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} & \frac{3}{4} + \frac{1}{4} \end{bmatrix} \tag{1.11}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2. \tag{1.12}$$

Thus L is orthogonal and the displayed L^{T} is its inverse.

1.6.2 The trigonometric pattern

The preceding example belongs to the family

$$Q(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}. \quad (1.13)$$

Multiplying by the transpose gives

$$\begin{aligned} Q(\theta)Q(\theta)^\top &= \begin{bmatrix} \sin^2 \theta + \cos^2 \theta & -\sin \theta \cos \theta + \cos \theta \sin \theta \\ -\cos \theta \sin \theta + \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix} \\ &= I_2. \end{aligned}$$

The identity $\sin^2 \theta + \cos^2 \theta = 1$ makes this family orthogonal for every real θ .

The numerical example corresponds to $L = Q(\pi/6)$, since $\sin(\pi/6) = 1/2$ and $\cos(\pi/6) = \sqrt{3}/2$. An equivalent parameterization uses cosine on the diagonal:

$$P(\varphi) = \begin{bmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{bmatrix}.$$

The relation $Q(\theta) = P(\pi/2 - \theta)$ follows from the complementary-angle identities. Thus the same numerical matrix is $L = P(\pi/3)$. In either parameterization, orthogonality follows from the same trigonometric identity. Such matrices provide a foundation for describing changes of orientation in robotics.

1.7 Key identities

For matrices of compatible dimensions, the operations developed in this chapter can be collected as follows:

Operation or property	Rule
Addition	$(M + N)_{ij} = m_{ij} + n_{ij}$
Multiplication	$(MN)_{ij} = \sum_{k=1}^n m_{ik}n_{kj}$
Transpose	$(M^\top)_{ij} = m_{ji}$
Transpose of a product	$(MN)^\top = N^\top M^\top$
Inverse	$MM^{-1} = M^{-1}M = I$
Invertibility of a square matrix	$\det(M) \neq 0$
Orthogonality	$M^{-1} = M^\top$

In the multiplication formula, n is the number of columns of M and the number of rows of N . Addition requires equal dimensions, whereas a product requires only that these inner dimensions match. Inverse and orthogonality conditions apply to square matrices. These distinctions determine which algebraic manipulations are valid before any numerical calculation begins.

1.8 Additional exercises

Show your calculations and justify each conclusion. Check dimensions before performing an operation. These exercises can be completed by hand.

1. **Dimensions and valid operations.** Let $A \in \mathbb{R}^{3 \times 2}$, $B \in \mathbb{R}^{2 \times 3}$, and $C \in \mathbb{R}^{3 \times 3}$. Determine which expressions are defined and give the dimensions of each valid result:

$$A + B, \quad AB, \quad BA, \quad CA, \quad AC, \quad A^T + C, \quad B + A^T.$$

2. **Addition and multiplication.** Consider

$$M = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Compute $M + N$, MN , NM , and Mv . Do M and N commute? Verify numerically that $(M + N)v = Mv + Nv$.

3. **Transposing a product.** Let

$$U = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 2 & 0 \end{bmatrix}.$$

Compute UV and $(UV)^T$, then independently compute $V^T U^T$ and compare the results. Is $U^T V^T$ defined? Explain why it cannot equal $(UV)^T$ in this example.

4. **Determinants and inverses.** For

$$D = \begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix},$$

use determinants to decide which matrices are invertible. Find D^{-1} . Verify that $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ is the inverse of K by multiplying in both orders.

5. **Testing orthogonality.** Consider

$$R = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Test each matrix for orthogonality using its transpose. For each orthogonal matrix, write its inverse. Is an invertible matrix necessarily orthogonal? Support your answer with one of these examples.

6. **A product of orthogonal matrices.** Suppose P and Q are real orthogonal $n \times n$ matrices. Use the transpose-of-a-product rule to show that $(PQ)(PQ)^T = I_n$. Deduce that PQ is orthogonal and express $(PQ)^{-1}$ in terms of P^T and Q^T , paying attention to their order.

Chapter 2

Reference frames and planar rotations

A robot must relate what its sensors observe to a description of the surrounding world. An obstacle may be directly ahead of the robot, yet lie in a different direction when described on a map. Neither description is ambiguous once the reference frame is specified. The task is to express the same geometric point in different coordinate systems.

This chapter develops that change of coordinates for planar frames with a common origin. Their relative orientation is described by one angle, or equivalently by a rotation matrix. The matrix formulation makes it possible to transform coordinates, reverse a transformation, and compose transformations through several frames using the linear algebra of Chapter 1.

2.1 Coordinates depend on a reference frame

A planar Cartesian reference frame consists of an origin and two perpendicular unit directions, denoted $\mathbf{x}$ and $\mathbf{y}$. A world frame W provides a fixed reference for a map, while a robot frame R is attached to the robot. A sensor can have its own frame S .

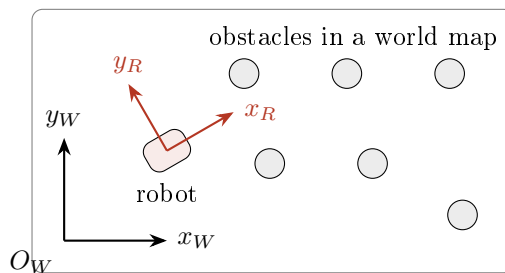


Figure 2.1: A map and a robot use different reference frames. In a general setting their origins and orientations both differ.

If the robot's x_R axis points forward and y_R points left, an obstacle one unit ahead has robot coordinates $[1\ 0]^T$. An obstacle one unit to the left has coordinates $[0\ 1]^T$. These descriptions alone do not specify where the obstacles belong on the world map; the relation between the frames is also needed.

2.1.1 Notation and the common-origin assumption

Let B denote a geometric point. Its coordinates are column vectors,

$$B^W = \begin{bmatrix} x_B^W \\ y_B^W \end{bmatrix}, \quad B^R = \begin{bmatrix} x_B^R \\ y_B^R \end{bmatrix}. \quad (2.1)$$

The superscript states the frame in which the coordinates are expressed. The point B does not change when its coordinate description changes.

For now, assume that W and R share an origin O and have the same orientation convention for their axes. Let $\theta = \theta_R^W$ be the signed angle from x_W to x_R , positive counterclockwise. Only a rotation separates the two frames. Figure 2.2 shows how projections onto their axes produce different coordinates for the same point.

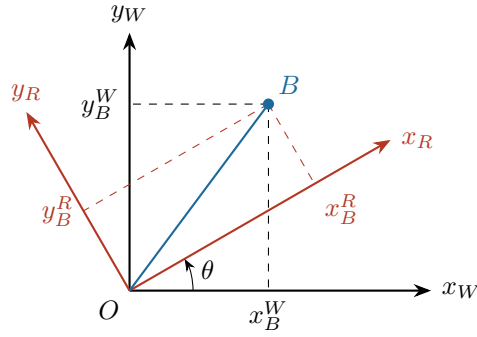


Figure 2.2: A point expressed in two frames with a common origin. Black axes belong to W and red axes to R ; dashed lines give the coordinate projections.

2.2 Constructing the rotation matrix

2.2.1 The columns are the rotated basis vectors

The direction $\mathbf{x}_R$ makes angle θ with x_W . Because it is a unit vector, its components in the world frame are

$$[\mathbf{x}_R]^W = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}. \quad (2.2)$$

The direction $\mathbf{y}_R$ is a further $\pi/2$ counterclockwise, so

$$[\mathbf{y}_R]^W = \begin{bmatrix} \cos(\theta + \pi/2) \\ \sin(\theta + \pi/2) \end{bmatrix} = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}. \quad (2.3)$$

Here brackets with a superscript mean the components of a geometric vector in the indicated frame. The negative sign in the first component of $[\mathbf{y}_R]^W$ is essential: for a small positive angle, $\mathbf{y}_R$ points to the left of y_W .

Putting these two column vectors side by side defines the rotation matrix:

$$R_R^W = [\mathbf{x}_R]^W \quad [\mathbf{y}_R]^W = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (2.4)$$

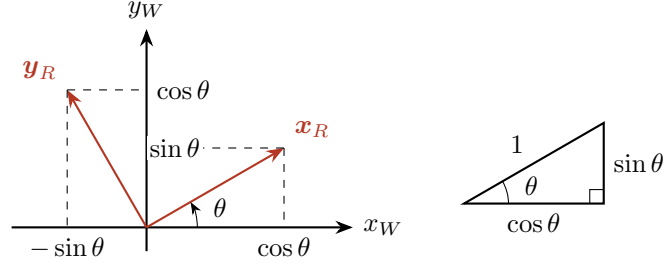


Figure 2.3: Components of the robot's unit basis vectors in the world frame. The right triangle explains the sine and cosine components of $\mathbf{x}_R$.

The subscript identifies the frame whose orientation is being described, and the superscript identifies the reference frame. As a coordinate map, R_R^W takes components expressed in R and returns components expressed in W . An angle is a compact description of orientation; the matrix provides a linear map on coordinate vectors once that angle is fixed.

2.2.2 Orthogonality

Write $c = \cos \theta$ and $s = \sin \theta$. Direct multiplication gives

$$(R_R^W)^\top R_R^W = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} = \begin{bmatrix} c^2 + s^2 & 0 \\ 0 & c^2 + s^2 \end{bmatrix} = I_2.$$

Thus R_R^W is orthogonal and

$$(R_R^W)^{-1} = (R_R^W)^\top = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (2.5)$$

Its determinant is $c^2 + s^2 = 1$. This last property distinguishes a rotation from an orthogonal reflection, whose determinant is -1 .

2.3 Transforming the coordinates of a point

The vector from the common origin to B can be decomposed along the robot axes:

$$\overrightarrow{OB} = x_B^R \mathbf{x}_R + y_B^R \mathbf{y}_R.$$

Expressing both basis vectors in W gives

$$\boxed{B^W = R_R^W B^R.} \quad (2.6)$$

In scalar form, this is

$$x_B^W = \cos \theta x_B^R - \sin \theta y_B^R, \quad (2.7)$$

$$y_B^W = \sin \theta x_B^R + \cos \theta y_B^R. \quad (2.8)$$

The dimensions are $(2 \times 2)(2 \times 1) = (2 \times 1)$, and the input frame R agrees with the matrix subscript. Both checks are useful: correct matrix dimensions alone do not establish that the intended frames match.

2.3.1 A geometric derivation

For $B \neq O$, let d be the distance from O to B , let α be the angle from x_R to $\overrightarrow{OB}$, and let β be the angle from x_W to that vector. The angles satisfy $\beta = \alpha + \theta$ (modulo 2π). Consequently,

$$x_B^R = d \cos \alpha, \quad y_B^R = d \sin \alpha, \quad x_B^W = d \cos \beta, \quad y_B^W = d \sin \beta.$$

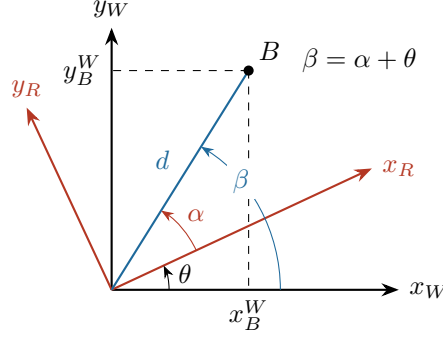


Figure 2.4: The same position vector has angle α in R and angle $\beta = \alpha + \theta$ in W . Its length d is independent of the frame.

Expanding the angle sums gives

$$\begin{aligned} x_B^W &= d \cos(\alpha + \theta) = d \cos \alpha \cos \theta - d \sin \alpha \sin \theta \\ &= \cos \theta x_B^R - \sin \theta y_B^R, \\ y_B^W &= d \sin(\alpha + \theta) = d \cos \alpha \sin \theta + d \sin \alpha \cos \theta \\ &= \sin \theta x_B^R + \cos \theta y_B^R. \end{aligned}$$

These are precisely the entries of (2.6). The matrix formula also applies at $B = O$, where both coordinate vectors are zero and a direction angle is unnecessary.

2.3.2 Worked example: a thirty-degree orientation

Suppose

$$B^R = \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix}, \quad \theta = \frac{\pi}{6}.$$

Then

$$B^W = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix} \tag{2.9}$$

$$= \begin{bmatrix} 0.4\sqrt{3} - 0.25 \\ 0.4 + 0.25\sqrt{3} \end{bmatrix} \approx \begin{bmatrix} 0.4428 \\ 0.8330 \end{bmatrix}. \tag{2.10}$$

The coordinates change, but the point's distance from the common origin does not. Indeed, orthogonality implies

$$(B^W)^\top B^W = (B^R)^\top (R_R^W)^\top R_R^W B^R = (B^R)^\top B^R = 0.8^2 + 0.5^2 = 0.89.$$

This is a change of coordinate description of a fixed point, rather than a physical motion of that point.

2.4 Reversing a coordinate transformation

To recover robot coordinates from world coordinates, multiply Equation (2.6) on the left by the inverse:

$$(R_R^W)^{-1} B^W = (R_R^W)^{-1} R_R^W B^R = B^R.$$

Using orthogonality,

$$\boxed{B^R = (R_R^W)^\top B^W = R_W^R B^W, \quad R_W^R = (R_R^W)^\top.} \quad (2.11)$$

Reversing the frame relation exchanges the frame indices and negates the relative angle. If $R(\theta)$ denotes the matrix in Equation (2.4), then $R(\theta)^\top = R(-\theta)$. For arbitrary frames A and F , the same rule reads

$$R_F^A = (R_A^F)^\top. \quad (2.12)$$

For example, applying the transpose to the exact vector in Equation (2.10) recovers $[0.8 \ 0.5]^\top$.

2.5 Composing rotations through several frames

A measurement may begin in a sensor frame S , pass through a robot frame R , and finally be expressed in W . Assume again that all origins coincide. Let the orientation of R relative to W be θ , and the orientation of S relative to R be φ .

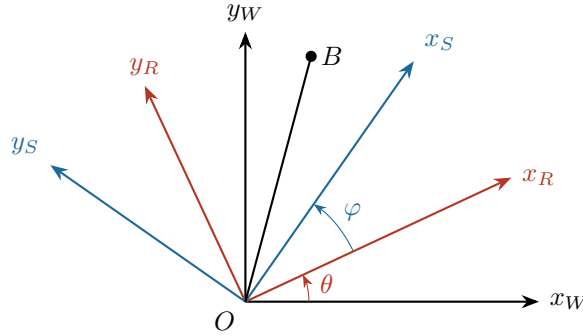


Figure 2.5: Three frames with a common origin. The sensor frame (blue) is rotated by φ relative to the robot frame (red), which is rotated by θ relative to the world frame (black).

The two successive coordinate transformations are

$$B^R = R_S^R B^S, \quad B^W = R_R^W B^R.$$

Substitution gives

$$B^W = R_R^W R_S^R B^S, \quad \boxed{R_S^W = R_R^W R_S^R.} \quad (2.13)$$

The rightmost matrix acts first. Its output is expressed in R , precisely the frame expected by the matrix to its left. In two dimensions,

$$R_R^W = R(\theta), \quad R_S^R = R(\varphi), \quad R_S^W = R(\theta + \varphi),$$

as can be verified with the sine and cosine addition identities.

2.5.1 Reading the frame indices

The general composition rule is

$$R_C^A = R_B^A R_C^B. \quad (2.14)$$

The subscript of the left factor matches the superscript of the next factor. The remaining indices identify the overall output and input frames. This is a rule about the meanings of the maps, not ordinary cancellation of numbers.

For example,

$$R_2^3 R_4^2 R_5^4 R_1^5 = R_1^3. \quad (2.15)$$

Reading from right to left, the coordinates pass through frames $1 \rightarrow 5 \rightarrow 4 \rightarrow 2 \rightarrow 3$. By contrast, in the written product $R_2^3 R_4^2$, the inner frame labels 2 and 4 do not match. Its factors cannot be composed as that sequence of coordinate maps merely by cancelling the repeated 3.

Planar rotation matrices happen to commute numerically: $R(\theta)R(\varphi) = R(\varphi)R(\theta)$. This special property does not change the input and output frame labels of an individual map. Keeping the frame chain explicit also prepares the notation for three-dimensional rotations, where commutativity does not hold in general.

2.5.2 Simplifying chains containing transposes

First apply the transpose rules, then inspect the frame chain. Consider

$$R_2^3 (R_2^4)^\top (R_5^1 R_4^5)^\top.$$

Transposing the last product reverses its factor order, so

$$\begin{aligned} R_2^3 (R_2^4)^\top (R_5^1 R_4^5)^\top &= R_2^3 R_4^2 (R_4^5)^\top (R_5^1)^\top \\ &= R_2^3 R_4^2 R_5^4 R_1^5 \\ &= R_1^3. \end{aligned}$$

Two different operations are involved: a transpose exchanges the frame indices of one rotation matrix, and transposing a product reverses the order of its factors. Both are necessary to obtain a consistent chain.

2.6 The limit of a rotation-only model

The formula $B^W = R_R^W B^R$ describes point coordinates when the origins coincide. In a general robot configuration, the origin O_R is displaced from O_W , as in Figure 2.6. A rotation matrix describes the relative orientation, but contains no information about this displacement.

To see the limitation, take the point $B = O_R$. Its robot coordinates are zero, so $R_R^W B^R = 0$ for every rotation matrix. Its world coordinates, however, are the nonzero displacement of O_R from O_W . A complete point-coordinate transformation must therefore include translation as well as rotation. The rotation rules remain applicable to direction vectors, which do not depend on the choice of origin. Distinguishing point positions from directions is essential when extending the model.

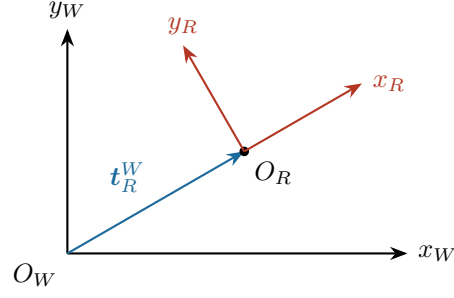


Figure 2.6: Different origins introduce a translation $\mathbf{t}_R^W$. Orientation alone cannot determine the world coordinates of a point measured relative to O_R .

2.7 Key identities

Operation	Identity
Planar rotation	$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
Point coordinates, common origin	$B^W = R_R^W B^R$
Reverse transformation	$R_W^R = (R_R^W)^\top = R(-\theta)$
Composition	$R_S^W = R_R^W R_S^R$
Planar angle composition	$R(\theta)R(\varphi) = R(\theta + \varphi)$

2.8 Additional exercises

Use column vectors, show your calculations, and label the input and output frames. Unless stated otherwise, all frames share an origin.

1. **Sketching frames and coordinates.** Draw world axes x_W, y_W and robot axes rotated counterclockwise by $\pi/2$. A point has $B^R = [2 \ 1]^\top$. Locate it on your sketch, compute B^W , and check that your answer agrees with its position relative to the world axes.
2. **Constructing a rotation matrix.** The robot frame is rotated by $-\pi/4$ relative to the world frame. Write $[x_R]^W$ and $[y_R]^W$, and use them to construct R_R^W . Verify $(R_R^W)^\top R_R^W = I_2$ and $\det(R_R^W) = 1$.
3. **Forward and reverse transformations.** Let $\theta_R^W = \pi/6$ and $B^R = [1 \ 2]^\top$. Find B^W in exact form. Construct R_W^R and use it to recover B^R . Verify that the squared distance from the common origin is the same in both coordinate descriptions.
4. **From a sensor to the world.** Suppose $\theta_R^W = \pi/6$, $\theta_S^R = \pi/3$, and $B^S = [2 \ -1]^\top$. Compute R_S^W and B^W . Check your result by first finding B^R and then transforming it to W . What is the orientation of S relative to W ?
5. **Frame chains and transposes.** Simplify the first two expressions to a single rotation matrix. For the third, explain why the written frame chain is inconsistent, even though the numerical matrix dimensions permit multiplication:

$$R_B^A R_C^B R_D^C, \quad (R_A^B)^\top (R_C^D R_B^C)^\top, \quad R_B^A R_D^C.$$

Assume distinct letters denote distinct frames. State the input and output frames of each valid chain.

6. **Why an origin matters.** The robot axes are parallel to the world axes, but the robot origin has world coordinates $[3 \ 1]^\top$. A point has robot coordinates $[2 \ 0]^\top$. Use a sketch to find its world coordinates. What would the rotation-only formula predict, and what information would it omit?
7. **Length preservation.** Let $v^W = R_R^W v^R$ be two coordinate descriptions of a direction vector. Using orthogonality, show that $(v^W)^\top v^W = (v^R)^\top v^R$. Explain geometrically why a change of reference frame cannot change the length of the vector.

Chapter 3

Planar rigid transformations and homogeneous coordinates

Rotation describes the relative orientation of two frames, but a mobile robot also changes position. Its origin generally differs from the origin of the world map. A sensor mounted on the robot may have yet another origin and orientation. Relating point coordinates between these frames requires both rotation and translation.

This chapter extends the rotation model of Chapter 2 to planar rigid transformations. The geometric construction leads first to an affine coordinate relationship, then to a homogeneous matrix representation that combines rotation and translation in a single multiplication. The same representation makes composition and inversion systematic.

3.1 Rotation and translation between two frames

Let O_W and O_R be the origins of world frame W and robot frame R . Define

$$p_R^W = \begin{bmatrix} p_{Rx}^W \\ p_{Ry}^W \end{bmatrix}$$

as the position of O_R relative to O_W , expressed in world coordinates. The rotation matrix R_R^W describes the orientation of R relative to W . Together, p_R^W and R_R^W specify the robot frame's planar pose.

For a point B , the geometric displacement vectors obey

$$\overrightarrow{O_W B} = \overrightarrow{O_W O_R} + \overrightarrow{O_R B}. \quad (3.1)$$

This triangle relation is independent of coordinates. To turn it into a numerical calculation, however, every vector being added must be expressed in the same frame.

The column B^R contains the components of $\overrightarrow{O_R B}$ along the robot axes. The column p_R^W uses the world axes. Adding these columns directly would combine components measured along different directions. First rotate B^R into world-aligned components; then add the translation:

$$\boxed{B^W = p_R^W + R_R^W B^R.} \quad (3.2)$$

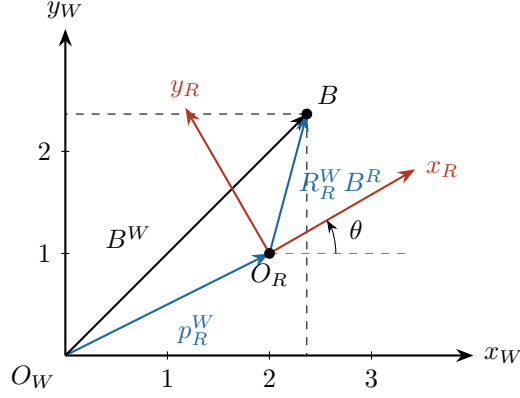


Figure 3.1: The displacement to B is the sum of the displacement to the robot origin and the displacement from that origin to B . All three arrow labels give components in W . The geometry uses $p_R^W = [2 \ 1]^T$, $\theta = \pi/6$, and $B^R = [1 \ 1]^T$.

Here $R_R^W B^R$ describes the displacement from O_R to B using world directions. It is not, by itself, the position of B relative to O_W .

3.1.1 An intermediate frame makes the geometry explicit

Introduce a frame A with origin $O_A = O_R$ and axes parallel to those of W . Frames R and A differ only by rotation, so the previous chapter gives

$$B^A = R_R^A B^R.$$

Since A and W have the same orientation, $R_R^A = R_R^W$. Moreover, the components in A use exactly the directions needed for addition to p_R^W . Thus $B^W = p_R^W + B^A$, recovering Equation (3.2).

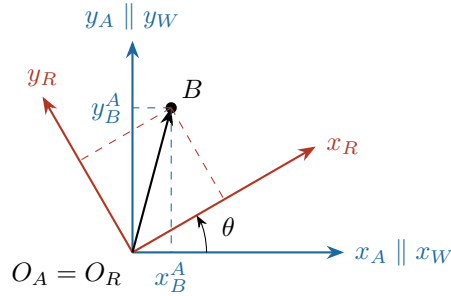


Figure 3.2: The auxiliary frame A shares the robot origin but is aligned with the world. Rotating from R to A resolves the displacement into components that can be added to the world position of O_R .

3.1.2 Why the relationship is affine

A map $f(b) = Rb + p$ is called affine: it consists of a linear map followed by an offset. With a nonzero translation, it is not a linear map of the original two coordinates, since $f(0) = p \neq 0$. In

the present setting R is a rotation, so the transformation is rigid: it preserves distances between points. Distances to the coordinate origin need not be preserved when the origin changes.

3.1.3 Worked example

Suppose the given quantities are

$$B^R = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ m}, \quad p_R^W = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ m}, \quad \theta_R^W = \frac{\pi}{6}.$$

The rotation matrix is dimensionless. Applying the affine relationship gives

$$B^W = \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (3.3)$$

$$= \begin{bmatrix} (3 + \sqrt{3})/2 \\ (3 + \sqrt{3})/2 \end{bmatrix} \approx \begin{bmatrix} 2.3660 \\ 2.3660 \end{bmatrix} \text{ m}. \quad (3.4)$$

The coordinate arithmetic uses metres throughout. The result agrees with Figure 3.1: the point lies above and slightly to the right of the robot origin. Keeping the exact expressions until the final step avoids rounding errors when reversing the calculation.

3.2 Reversing the affine relationship

Suppose B^W is known and B^R is required. Subtract the world-coordinate translation before applying the inverse rotation:

$$B^W - p_R^W = R_R^W B^R, \quad (3.5)$$

$$\boxed{B^R = (R_R^W)^\top (B^W - p_R^W)}. \quad (3.6)$$

The difference in parentheses is expressed in W ; the transpose maps it to R . The parentheses matter: the rotation acts on both terms, not just on B^W . Equivalently,

$$B^R = -(R_R^W)^\top p_R^W + (R_R^W)^\top B^W. \quad (3.7)$$

This has the same affine structure as the forward relationship, with

$$R_W^R = (R_R^W)^\top, \quad p_W^R = -(R_R^W)^\top p_R^W. \quad (3.8)$$

The reversed translation is not generally just $-p_R^W$: it must also be expressed along the axes of the reverse transformation's output frame.

For the exact coordinates in Equation (3.4),

$$B^W - p_R^W = \begin{bmatrix} (\sqrt{3} - 1)/2 \\ (1 + \sqrt{3})/2 \end{bmatrix} \text{ m}.$$

Multiplying by $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$ recovers $B^R = [1 \ 1]^\top \text{ m}$. If the rounded value $[2.37 \ 2.37]^\top$ is used instead, the recovered result is approximately $[1.0054 \ 1.0015]^\top \text{ m}$, rather than exactly $[1 \ 1]^\top$.

3.3 Composing transformations through a sensor frame

A sensor frame S may be displaced and rotated relative to the robot frame. Given the sensor's pose in R and the robot's pose in W , the two coordinate relationships are

$$B^R = p_S^R + R_S^R B^S, \quad B^W = p_R^W + R_R^W B^R.$$

Substitution yields

$$B^W = p_R^W + R_R^W (p_S^R + R_S^R B^S) \quad (3.9)$$

$$= \underbrace{p_R^W + R_R^W p_S^R}_{p_S^W} + \underbrace{R_R^W R_S^R}_{R_S^W} B^S. \quad (3.10)$$

The overall rotation and translation are therefore

$$\boxed{R_S^W = R_R^W R_S^R, \quad p_S^W = p_R^W + R_R^W p_S^R.} \quad (3.11)$$

The sensor translation must be rotated into W before it can be added to the robot translation. Frame labels explain why the expression is valid and why simply adding $p_R^W + p_S^R$ would be incorrect in general.

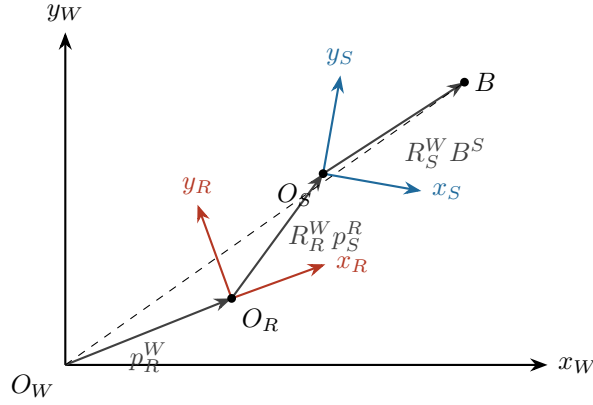


Figure 3.3: A world–robot–sensor chain with distinct origins. Each displacement along the path to B is labeled in world components so that the arrows can be added numerically. The dashed line is the direct displacement to B .

Repeated substitution works for any number of frames, but each additional frame introduces another translation term. A representation that absorbs these additions into matrix multiplication makes longer chains easier to manage.

3.4 Homogeneous coordinates and transformation matrices

3.4.1 Adding one coordinate

For a point with coordinates B^R , define its homogeneous representation by appending a dimensionless one:

$$\tilde{B}^R = \begin{bmatrix} B^R \\ 1 \end{bmatrix} = \begin{bmatrix} x_B^R \\ y_B^R \\ 1 \end{bmatrix}. \quad (3.12)$$

The additional coordinate is bookkeeping for translation, not a third spatial coordinate. Define the planar homogeneous transformation matrix as

$$H_R^W = \begin{bmatrix} R_R^W & p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix}. \quad (3.13)$$

Its blocks have dimensions 2×2 , 2×1 , 1×2 , and 1×1 , respectively. Written entry by entry, it is

$$H_R^W = \begin{bmatrix} \cos \theta & -\sin \theta & p_{Rx}^W \\ \sin \theta & \cos \theta & p_{Ry}^W \\ 0 & 0 & 1 \end{bmatrix}.$$

The upper-left block contains dimensionless rotation coefficients; the first two entries of the last column carry the chosen length unit. Block divisions are visual aids, not extra mathematical operations.

3.4.2 One product performs the full transformation

Block multiplication gives

$$H_R^W \tilde{B}^R = \begin{bmatrix} R_R^W & p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix} \begin{bmatrix} B^R \\ 1 \end{bmatrix} = \begin{bmatrix} R_R^W B^R + p_R^W \\ 1 \end{bmatrix} = \tilde{B}^W. \quad (3.14)$$

Thus the affine transformation becomes a single matrix multiplication:

$$\tilde{B}^W = H_R^W \tilde{B}^R. \quad (3.15)$$

The first two output entries are the desired point coordinates. For the normalized point representation used here, the last entry remains one. The fixed bottom row $[0 \ 0 \ 1]$ is what preserves this property.

For the running example, using metres for the position entries,

$$\begin{bmatrix} \sqrt{3}/2 & -1/2 & 2 \\ 1/2 & \sqrt{3}/2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (3 + \sqrt{3})/2 \\ (3 + \sqrt{3})/2 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 2.3660 \\ 2.3660 \\ 1 \end{bmatrix}.$$

The physical point coordinates are the first two entries in metres; the final one has no length unit.

3.5 Composition by matrix multiplication

Using homogeneous coordinates for the sensor chain,

$$\tilde{B}^R = H_S^R \tilde{B}^S, \quad \tilde{B}^W = H_R^W \tilde{B}^R.$$

Combining the two maps gives

$$H_S^W = H_R^W H_S^R, \quad \tilde{B}^W = H_R^W H_S^R \tilde{B}^S. \quad (3.16)$$

Multiplying the blocks verifies that this contains both composition rules:

$$H_R^W H_S^R = \begin{bmatrix} R_R^W & p_R^W \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_S^R & p_S^R \\ 0 & 1 \end{bmatrix} \quad (3.17)$$

$$= \begin{bmatrix} R_R^W R_S^R & p_R^W + R_R^W p_S^R \\ 0 & 1 \end{bmatrix} = H_S^W. \quad (3.18)$$

Here each 0 in the bottom-left block denotes a 1×2 zero row. The product is another homogeneous rigid transformation with bottom row $[0 \ 0 \ 1]$. As before, the rightmost map acts first and adjacent frame indices must match. For example,

$$H_1^2 H_3^1 H_4^3 H_5^4 = H_5^2$$

maps coordinates through $5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$.

Unlike pure planar rotations, planar transformations with translation do not generally commute. The term $R_R^W p_S^R$ shows why: a preceding rotation changes the direction in which a subsequent local translation acts.

3.6 Inverting a homogeneous transformation

3.6.1 Why the transpose is insufficient

Expanding the determinant along the bottom row gives

$$\det(H_R^W) = \det(R_R^W) = 1.$$

Every matrix of this rigid-transformation form is therefore invertible. However, it is generally not orthogonal. If its translation is nonzero, its transpose has bottom row $[(p_R^W)^\top \ 1]$ rather than $[0 \ 0 \ 1]$. Consequently, transposing the full matrix does not produce its inverse. When the translation is zero, the full matrix is orthogonal and the transpose does coincide with the inverse.

3.6.2 Deriving the inverse from the affine relationship

Equation (3.7) already identifies the reverse rotation and translation. Place these in their corresponding blocks:

$$H_W^R = (H_R^W)^{-1} = \begin{bmatrix} (R_R^W)^\top & -(R_R^W)^\top p_R^W \\ 0_{1 \times 2} & 1 \end{bmatrix}. \quad (3.19)$$

This gives $\tilde{B}^R = H_W^R \tilde{B}^W$ without a general-purpose 3×3 inversion. Transpose the rotation block, compute the rotated negative translation, and retain the bottom row.

For a compact verification, write $R = R_R^W$ and $p = p_R^W$. Then

$$\begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R^\top & -R^\top p \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R R^\top & -R R^\top p + p \\ 0 & 1 \end{bmatrix} = I_3.$$

The reverse product also equals I_3 , since its translation block is $R^\top p - R^\top p = 0$.

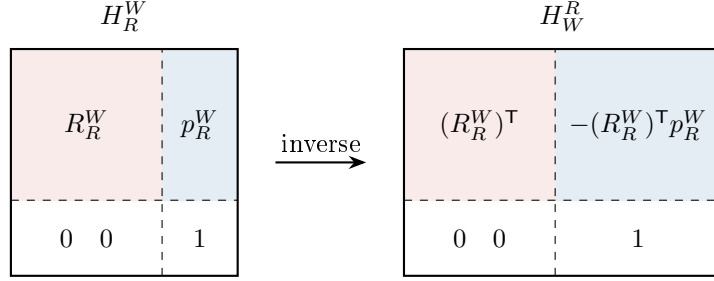


Figure 3.4: Block structure of a transform and its inverse. Red blocks contain rotation; blue blocks contain translation. The inverse translation must be rotated as well as negated. Block widths are schematic.

3.6.3 Worked inverse

For $p_R^W = [2 \ 1]^T$ m and $\theta_R^W = \pi/6$,

$$\begin{aligned} -(R_R^W)^T p_R^W &= - \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -\sqrt{3} - 1/2 \\ 1 - \sqrt{3}/2 \end{bmatrix} \approx \begin{bmatrix} -2.2321 \\ 0.1340 \end{bmatrix} \text{ m.} \end{aligned}$$

Hence, with translation entries in metres,

$$H_W^R = \begin{bmatrix} \sqrt{3}/2 & 1/2 & -\sqrt{3} - 1/2 \\ -1/2 & \sqrt{3}/2 & 1 - \sqrt{3}/2 \\ 0 & 0 & 1 \end{bmatrix}. \quad (3.20)$$

Its last column locates the world origin in robot coordinates. Its upper-left block describes the world axes in the robot frame. These meanings provide geometric checks on the inverse, in addition to verifying the matrix product.

3.7 Key identities

Operation	Identity
Forward point transformation	$B^W = p_R^W + R_R^W B^R$
Reverse point transformation	$B^R = (R_R^W)^T (B^W - p_R^W)$
Homogeneous point transformation	$\tilde{B}^W = H_R^W \tilde{B}^R$
Composition	$H_S^W = H_R^W H_S^R$
Reverse rotation	$R_W^R = (R_R^W)^T$
Reverse translation	$p_W^R = -(R_R^W)^T p_R^W$

Always specify the frame of a translation, express displacements in a common frame before adding them, and distinguish the inverse of a rotation matrix from the inverse of a full homogeneous transformation.

3.8 Additional exercises

Use metres for all position coordinates and counterclockwise-positive angles. Show your calculations and keep exact values until the final numerical step.

1. **A translated and rotated robot.** Let $p_R^W = [3 \ 1]^T$, $\theta_R^W = \pi/2$, and $B^R = [2 \ -1]^T$. Sketch the two frames and the point. Compute B^W using the affine relationship and explain why $p_R^W + B^R$ gives a different result.
2. **Recovering robot coordinates.** Suppose $p_R^W = [1 \ -2]^T$, $\theta_R^W = -\pi/2$, and $B^W = [4 \ -3]^T$. Find B^R . Substitute it into the forward relationship to verify your answer. State the frame of every vector being added.
3. **Building a homogeneous matrix.** Construct H_R^W for $p_R^W = [2 \ 0]^T$ and $\theta_R^W = \pi/4$. Use it to transform $B^R = [\sqrt{2} \ 0]^T$. Identify the physical coordinates in the output and explain the role of its last entry.
4. **World-robot-sensor composition.** Let $p_R^W = [2 \ 1]^T$, $\theta_R^W = \pi/2$, $p_S^R = [1 \ 0]^T$, and $\theta_S^R = -\pi/2$. Find $H_S^W = H_R^W H_S^R$ and use it to transform $B^S = [1 \ 2]^T$. Check your result by applying the two affine transformations separately.
5. **Inversion without a general matrix inverse.** For

$$H_R^W = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix},$$

compute H_W^R using its rotation and translation blocks. Verify $H_W^R H_R^W = I_3$. Compare your result with $(H_R^W)^T$ and identify why the transpose fails.

6. **Order of operations.** Consider the two maps, expressed using a fixed coordinate system,

$$T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Compute QT and TQ and apply each to $[0 \ 0 \ 1]^T$. Explain geometrically why rotating after translating differs from translating after rotating.

7. **Distances between points.** Two points obey $B^W = p + RB^R$ and $C^W = p + RC^R$, with R orthogonal. Show that $B^W - C^W = R(B^R - C^R)$ and use this to prove that their distance is preserved. Why need the distance of a single point from its coordinate origin not be preserved?

Chapter 4

Three-dimensional rotations and orientation representations

A robot that moves outside a plane requires a spatial description of orientation. One angle no longer suffices: a body can turn about three independent directions. The construction of a rotation matrix remains the same, however. Its columns are the unit axes of one frame expressed in another. In three dimensions, these columns have three components, giving a 3×3 matrix.

This chapter constructs rotations about individual axes, combines them into a yaw–pitch–roll sequence, recovers those angles from a rotation matrix, and introduces unit quaternions as a practical orientation representation. All frames used for point-coordinate transformations here share an origin; translation is set aside so that the orientation conventions can be developed explicitly.

4.1 Spatial frames and rotation matrices

Let W be a right-handed world frame and R a right-handed robot frame. Each has three mutually perpendicular unit axes. Write the robot’s axes in world components as $[\mathbf{x}_R]^W$, $[\mathbf{y}_R]^W$, and $[\mathbf{z}_R]^W$. Then

$$R_R^W = \begin{bmatrix} [\mathbf{x}_R]^W & [\mathbf{y}_R]^W & [\mathbf{z}_R]^W \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (4.1)$$

As in the planar case, the subscript gives the input frame and the superscript gives the output frame of the coordinate map. For a point B and common origins,

$$B^W = R_R^W B^R, \quad B^R = \begin{bmatrix} x_B^R \\ y_B^R \\ z_B^R \end{bmatrix}. \quad (4.2)$$

The same rotation maps the components of a direction vector between frames regardless of the locations of their origins.

Positive elementary angles follow the right-hand rule: point the right thumb along the positive rotation axis and curl the fingers in the positive direction. Viewed from the positive end of that axis toward the origin, the positive rotation appears counterclockwise. A perspective drawing may distort lengths and visible angles; it does not change this convention.

4.2 Elementary rotations

An elementary rotation leaves one axis fixed and rotates the perpendicular plane. The three basic cases are shown in Figure 4.1. Each is constructed by expressing the rotated basis vectors in the original frame, then placing those vectors in columns.

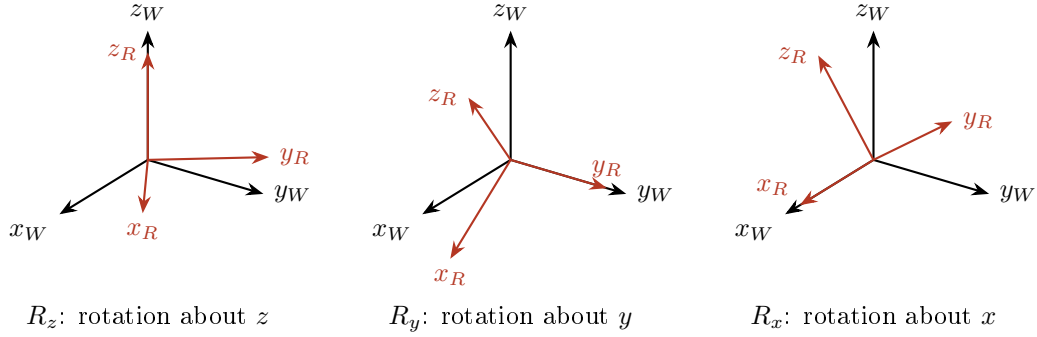


Figure 4.1: Elementary rotations in perspective. Black axes are the world frame; red axes are the rotated frame. The rotation axis is shared by the two frames. The three panels are separate constructions, each with a positive angle.

4.2.1 Rotation about the z axis

Let ψ denote a rotation about z_W . The z axis remains unchanged, while the x and y axes rotate in the xy plane:

$$[\mathbf{x}_R]^W = \begin{bmatrix} \cos \psi \\ \sin \psi \\ 0 \end{bmatrix}, \quad [\mathbf{y}_R]^W = \begin{bmatrix} -\sin \psi \\ \cos \psi \\ 0 \end{bmatrix}, \quad [\mathbf{z}_R]^W = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$R_z(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (4.3)$$

The upper-left block is exactly the planar rotation matrix. The last coordinate is unchanged because the rotation takes place in the plane perpendicular to z .

4.2.2 Rotation about the y axis

For a positive angle θ about y_W , the x axis turns toward negative z and the z axis turns toward positive x . The basis columns are

$$[\mathbf{x}_R]^W = \begin{bmatrix} \cos \theta \\ 0 \\ -\sin \theta \end{bmatrix}, \quad [\mathbf{y}_R]^W = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [\mathbf{z}_R]^W = \begin{bmatrix} \sin \theta \\ 0 \\ \cos \theta \end{bmatrix}.$$

Thus

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}. \quad (4.4)$$

The signs follow from the right-hand rule. The two-dimensional pattern is visible in the ordered plane (z, x) , where positive rotation carries z toward x ; using the order (x, z) without accounting for orientation would reverse the signs.

4.2.3 Rotation about the x axis

For a positive angle φ about x_W , the x axis is fixed and the y axis turns toward positive z . Hence

$$[\mathbf{x}_R]^W = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [\mathbf{y}_R]^W = \begin{bmatrix} 0 \\ \cos \varphi \\ \sin \varphi \end{bmatrix}, \quad [\mathbf{z}_R]^W = \begin{bmatrix} 0 \\ -\sin \varphi \\ \cos \varphi \end{bmatrix},$$

and

$$R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \varphi & -\sin \varphi \\ 0 & \sin \varphi & \cos \varphi \end{bmatrix}. \quad (4.5)$$

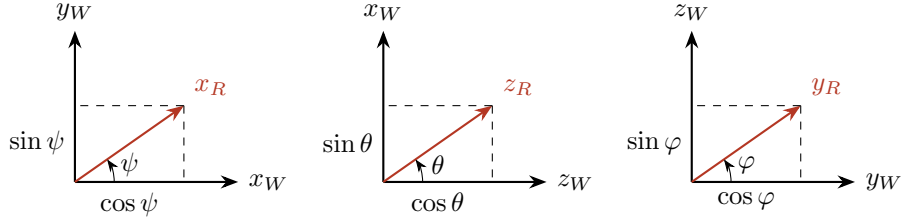


Figure 4.2: Plane views of the elementary rotations. From left to right, the ordered planes are (x_W, y_W) , (z_W, x_W) , and (y_W, z_W) . Each red vector has unit length; its projections explain the sine and cosine entries.

A quick sign check is to use a quarter-turn: $R_z(\pi/2)$ maps the first basis vector to the second, $R_y(\pi/2)$ maps the third to the first, and $R_x(\pi/2)$ maps the second to the third. At zero angle, all three matrices reduce to I_3 .

4.3 What makes a matrix a valid spatial rotation?

Write $R = [u \ v \ w]$. Its columns must be unit vectors and mutually perpendicular:

$$u^\top u = v^\top v = w^\top w = 1, \quad u^\top v = u^\top w = v^\top w = 0. \quad (4.6)$$

Together these six scalar constraints give $R^\top R = I_3$. They constrain the nine matrix entries to three continuous degrees of freedom. To describe a rotation between right-handed frames, rather than a reflection, one also requires $\det R = +1$. The determinant condition selects the proper-rotation component; it does not remove another continuous degree of freedom.

For a valid rotation,

$$R^{-1} = R^T, \quad R_F^A = (R_A^F)^T. \quad (4.7)$$

Composition still follows the frame chain, $R_C^A = R_B^A R_C^B$, and lengths are preserved. These are the same structural properties used in two dimensions.

4.3.1 An illustrative set of components

To illustrate how columns are assembled, consider the rough component array

$$M = \begin{bmatrix} .8 & -.3 & .4 \\ .2 & .7 & .15 \\ -.1 & .2 & .7 \end{bmatrix}. \quad (4.8)$$

Its first column describes a direction with a large positive x component, a smaller positive y component, and a negative z component. The other columns can be read in the same way. However, these illustrative numbers do *not* define a rotation matrix. For example,

$$u^T u = .8^2 + .2^2 + (-.1)^2 = .69 \neq 1, \quad u^T v = .8(-.3) + .2(.7) + (-.1)(.2) = -.12 \neq 0.$$

Even normalizing each column separately would not make the columns mutually perpendicular. A valid frame must satisfy all the orthonormality conditions and the right-handedness condition, not just have plausible-looking components.

4.4 Roll, pitch, and yaw as a sequence of rotations

A general spatial orientation can be represented by a sequence of three elementary rotations. Many sequences are possible, and the convention must be specified before interpreting three angle values. Here the convention is the *intrinsic Z-Y-X sequence*: each successive rotation is about an axis of the frame obtained after the previous rotation.

Starting from frame W , define intermediate frames 1 and 2 and final frame 3:

1. Rotate by yaw ψ about z_W to obtain frame 1.
2. Rotate by pitch θ about the new axis y_1 to obtain frame 2.
3. Rotate by roll φ about the new axis x_2 to obtain frame 3.

Although the angles are often listed as roll, pitch, yaw, the construction above proceeds through yaw, pitch, then roll.

The elementary frame relationships are

$$R_1^W = R_z(\psi), \quad R_2^1 = R_y(\theta), \quad R_3^2 = R_x(\varphi).$$

Applying the frame-chain rule gives

$$\boxed{R_3^W = R_1^W R_2^1 R_3^2 = R_z(\psi) R_y(\theta) R_x(\varphi).} \quad (4.9)$$

This product maps coordinates from frame 3 to W . When it multiplies B^3 , the rightmost factor acts first: $B^3 \rightarrow B^2 \rightarrow B^1 \rightarrow B^W$. This algebraic order is consistent with constructing the frames successively about their moving axes; it should not be confused with three rotations about fixed world axes in the same chronological order.

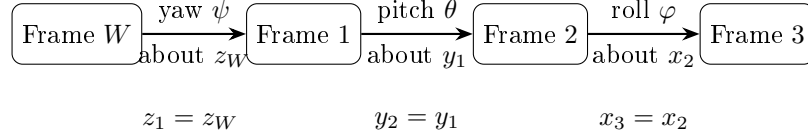


Figure 4.3: The intrinsic Z–Y–X construction. The axis used at each stage stays fixed during that stage. Arrows show the sequence of frame construction; coordinate maps from frame 3 back to W act in the reverse traversal.

4.4.1 Order matters in three dimensions

Unlike planar rotations, rotations about different spatial axes do not generally commute. With $e_x = [1 \ 0 \ 0]^\top$,

$$R_z(\pi/2)R_x(\pi/2)e_x = e_y, \quad R_x(\pi/2)R_z(\pi/2)e_x = e_z.$$

The first product leaves e_x unchanged in its first step, then turns it toward e_y . The second turns it toward e_y first, then toward e_z . Thus changing the factor order generally changes the orientation.

4.4.2 Expanded form

For compactness, write $c_a = \cos a$ and $s_a = \sin a$. Multiplying the three elementary matrices yields

$$R = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\varphi - s_\psi c_\varphi & c_\psi s_\theta c_\varphi + s_\psi s_\varphi \\ s_\psi c_\theta & s_\psi s_\theta s_\varphi + c_\psi c_\varphi & s_\psi s_\theta c_\varphi - c_\psi s_\varphi \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi \end{bmatrix}. \quad (4.10)$$

The factored form is often easier to evaluate and check. The expanded form is particularly useful for identifying which entries contain the information needed to recover the angles.

4.5 Worked example

Let the roll, pitch, and yaw angles be

$$\varphi = \frac{\pi}{4}, \quad \theta = \frac{\pi}{3}, \quad \psi = \frac{\pi}{2}.$$

Then

$$R_3^W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & \sqrt{3}/2 \\ 0 & 1 & 0 \\ -\sqrt{3}/2 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \quad (4.11)$$

$$= \begin{bmatrix} 0 & -\sqrt{2}/2 & \sqrt{2}/2 \\ 1/2 & \sqrt{6}/4 & \sqrt{6}/4 \\ -\sqrt{3}/2 & \sqrt{2}/4 & \sqrt{2}/4 \end{bmatrix}. \quad (4.12)$$

For instance, the first column has squared length $1/4 + 3/4 = 1$ and has zero dot product with the second column. More generally, the product of these three proper rotation matrices is itself a proper rotation matrix.

If $B^3 = [1 \ 0 \ 0]^\top$, its world coordinates are the first column of R_3^W . To return to frame 3, use $(R_3^W)^\top$. Thus the orientation representation serves directly as a coordinate transformation.

4.6 Recovering roll, pitch, and yaw

Let a valid rotation matrix be written as

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}.$$

For the convention in Equation (4.9), choose the principal pitch range $-\pi/2 < \theta < \pi/2$. In this nonsingular range,

$$\begin{cases} \varphi = \text{atan2}(r_{32}, r_{33}), \\ \theta = \text{atan2}\left(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}\right), \\ \psi = \text{atan2}(r_{21}, r_{11}). \end{cases} \quad (4.13)$$

The entries in Equation (4.10) explain these formulas:

$$(r_{32}, r_{33}) = c_\theta(s_\varphi, c_\varphi), \quad (r_{21}, r_{11}) = c_\theta(s_\psi, c_\psi).$$

Since $c_\theta > 0$ on the chosen range, each pair has the same direction as the corresponding sine–cosine pair. Also,

$$-r_{31} = s_\theta, \quad \sqrt{r_{32}^2 + r_{33}^2} = |c_\theta| = c_\theta.$$

Applied to Equation (4.12), the formulas give

$$\varphi = \text{atan2}(\sqrt{2}/4, \sqrt{2}/4) = \pi/4, \quad \theta = \text{atan2}(\sqrt{3}/2, 1/2) = \pi/3,$$

and $\psi = \text{atan2}(1/2, 0) = \pi/2$, recovering the original angles.

4.6.1 Why atan2 takes two arguments

The function $\text{atan2}(y, x)$ returns the angle of the pair (x, y) , using both signs to identify the quadrant. The argument order here is *vertical component first, horizontal component second*. The numeral 2 is part of the function’s name, not a multiplication or exponent.

The ratio y/x alone loses information: $(1, 1)$ and $(-1, -1)$ both have ratio one, but point in opposite directions. Ordinary $\arctan(y/x)$ returns the same principal angle for both, whereas

$$\text{atan2}(1, 1) = \pi/4, \quad \text{atan2}(-1, -1) = -3\pi/4.$$

Unlike division followed by arctangent, atan2 also handles $x = 0$ with $y \neq 0$, giving $\pi/2$ or $-\pi/2$.

Tangent has period π , while a full revolution has length 2π . The principal arctangent takes values in $(-\pi/2, \pi/2)$; a conventional principal range for atan2 spans 2π , such as $(-\pi, \pi]$. At the negative horizontal axis, π and $-\pi$ represent the same direction; endpoint handling can vary between implementations. At $(x, y) = (0, 0)$ there is no geometric direction, whatever numerical value a software routine returns.

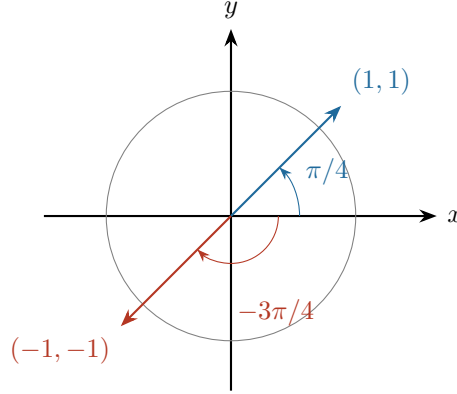


Figure 4.4: Equal ratios do not imply equal directions. The signs of both components distinguish the first and third quadrants. The circle indicates direction only; the two labeled vectors need not have unit length.

4.6.2 Singular orientations and angle ranges

The three-angle representation covers all orientations, but is not globally unique or nonsingular. At $\theta = \pm\pi/2$, the pairs used to extract yaw and roll both vanish:

$$r_{32} = r_{33} = r_{21} = r_{11} = 0.$$

Consequently, Equation (4.13) cannot determine yaw and roll separately. At $\theta = \pi/2$, for example, the matrix depends on their difference $\varphi - \psi$. This is the Euler-angle singularity often called *gimbal lock*. It is a limitation of the angle coordinates, not a loss of information in the rotation matrix.

An implementation must choose a convention for the singular case rather than treat $\text{atan2}(0, 0)$ as a uniquely meaningful angle. Near this case, small matrix perturbations can produce large changes in extracted yaw and roll. Away from it, the principal ranges are typically roll and yaw in a 2π interval, and pitch in $[-\pi/2, \pi/2]$.

4.7 Choosing and maintaining an orientation representation

Angles make individual turns easy to describe, while matrices make coordinate transformation and composition direct. The map $B^W = R_R^W B^R$ is linear in the coordinates of B for a fixed rotation; the dependence of R_R^W on the angles remains trigonometric.

Finite-precision arithmetic can cause repeatedly multiplied rotation matrices to drift from orthogonality. Useful checks are whether $R^T R$ remains close to I_3 and whether $\det R$ remains close to $+1$. Restoring unit column lengths alone is insufficient: perpendicularity and handedness must also be retained. A correction procedure must restore a proper orthonormal frame.

Angle representations have their own difficulties, including wrapping at the ends of the chosen intervals and singular orientations. Unit quaternions provide another representation used in robotics software. No representation removes the need to specify conventions and manage numerical error.

4.7.1 Unit quaternions in practice

A unit quaternion represents orientation using four real components. With the scalar component first, write

$$q = [q_w \ q_x \ q_y \ q_z]^\top, \quad q^\top q = q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1. \quad (4.14)$$

Four components with one unit-norm constraint leave three continuous degrees of freedom, just as for a rotation matrix. The quaternions q and $-q$ represent the same orientation; this discrete ambiguity does not add a degree of freedom. Component order must always be checked when exchanging data: some software places the scalar component last.

For a nonzero quaternion affected by small numerical drift, restoring unit norm is straightforward:

$$q \leftarrow \frac{q}{\|q\|}, \quad \|q\| = \sqrt{q_w^2 + q_x^2 + q_y^2 + q_z^2}.$$

A zero quaternion cannot be normalized and does not represent an orientation. Normalization enforces the representation's constraint; it does not undo every numerical error in the represented orientation.

The four components are less immediately intuitive than roll, pitch, and yaw, but are convenient for storing and exchanging orientations. A useful workflow is to retain a unit quaternion in software, convert to a rotation matrix for coordinate calculations, and extract angles when interpreting the result. To compose two orientations using familiar matrix algebra, convert both to matrices, multiply in the appropriate frame order, and convert back if needed. Direct quaternion composition is also possible and efficient; its algebra is outside the scope of this introduction.

Appendix A gives all six conversion directions between matrices, RPY angles, and scalar-first unit quaternions, including their validity conditions. For example, a quaternion can be converted to RPY angles to inspect yaw, while remembering that the chosen angle representation still has its pitch singularities.

4.8 Key identities

Property or operation	Rule
Proper rotation	$R^\top R = I_3, \det R = 1$
Coordinate transformation, common origin	$B^W = R_R^W B^R$
Reverse transformation	$R_W^R = (R_R^W)^\top$
Frame composition	$R_C^A = R_B^A R_C^B$
Intrinsic Z–Y–X convention	$R = R_z(\psi) R_y(\theta) R_x(\varphi)$
Nonsingular principal pitch range	$-\pi/2 < \theta < \pi/2$

4.9 Additional exercises

Use right-handed frames, column vectors, and the intrinsic Z–Y–X convention. Show your calculations and keep exact trigonometric values where possible.

1. **Elementary rotations.** Write $R_z(\pi/2)$, $R_y(\pi/2)$, and $R_x(\pi/2)$. For each matrix, identify its unchanged axis and determine the images of all three standard basis vectors. Check the signs using the right-hand rule.
2. **Checking a candidate matrix.** For the matrix M in Equation (4.8), compute the squared lengths of its three columns and all three pairwise dot products. Explain why normalizing the columns individually cannot produce a valid rotation matrix.
3. **A valid spatial orientation.** Consider

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B^R = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Compute $B^W = RB^R$, recover B^R using the transpose, and verify length preservation. State which elementary rotation R represents.

4. **Noncommutativity.** Compute $R_x(\pi/2)R_y(\pi/2)$ and $R_y(\pi/2)R_x(\pi/2)$. Apply both to $e_z = [0 \ 0 \ 1]^T$. Explain why the two results differ.
5. **From angles to a matrix and back.** Take $\varphi = \pi/4$, $\theta = 0$, and $\psi = \pi/2$. Compute $R = R_z(\psi)R_y(\theta)R_x(\varphi)$, then recover all three angles with Equation (4.13). Identify the frame axis used at each stage of the intrinsic construction.
6. **Quadrants and atan2.** Evaluate $\text{atan2}(1, -1)$, $\text{atan2}(-1, -1)$, and $\text{atan2}(-1, 0)$. Sketch the corresponding directions. Explain what goes wrong if each result is computed using only $\arctan(y/x)$.
7. **A singular orientation.** Set $\theta = \pi/2$ in Equation (4.10). Show that the result depends on $\varphi - \psi$. Verify that the triples $(\varphi, \theta, \psi) = (0, \pi/2, 0)$ and $(\pi/4, \pi/2, \pi/4)$ give the same matrix. Why can the extraction formulas not recover the two triples as distinct orientations?

Chapter 5

Three-dimensional rigid transformations and pose

A spatial reference frame can differ from another in both its origin and its orientation. Combining translation with the rotations developed in Chapter 4 gives a description of the pose of a rigid body. The geometry follows the planar construction: rotate a point's coordinates into the destination frame, then add the displacement between origins. The vectors now have three components, and homogeneous transformations have four rows and four columns.

5.1 Combining spatial rotation and translation

Let W be a world frame and R a frame attached to a rigid body. Define

$$p_R^W \in \mathbb{R}^3, \quad R_R^W \in \mathbb{R}^{3 \times 3},$$

where p_R^W is the position of the origin O_R expressed in W , and R_R^W maps vector components from R to W . The matrix is a proper rotation: $(R_R^W)^\top R_R^W = I_3$ and $\det R_R^W = 1$.

If a sensor reports a point B using coordinates B^R , its world coordinates are obtained by adding two displacements expressed in the same frame:

$$\boxed{B^W = R_R^W B^R + p_R^W.} \quad (5.1)$$

The first term describes the displacement from O_R to B in world components; the second locates O_R relative to O_W .

The dimensions provide a useful check:

$$\underbrace{B^W}_{3 \times 1} = \underbrace{R_R^W}_{3 \times 3} \underbrace{B^R}_{3 \times 1} + \underbrace{p_R^W}_{3 \times 1}.$$

Rotation alone maps direction vectors between frames. Translation is required for point coordinates because the two origins can differ. For example, subtracting the transformed coordinates of two points cancels p_R^W , leaving only the rotation of their displacement.

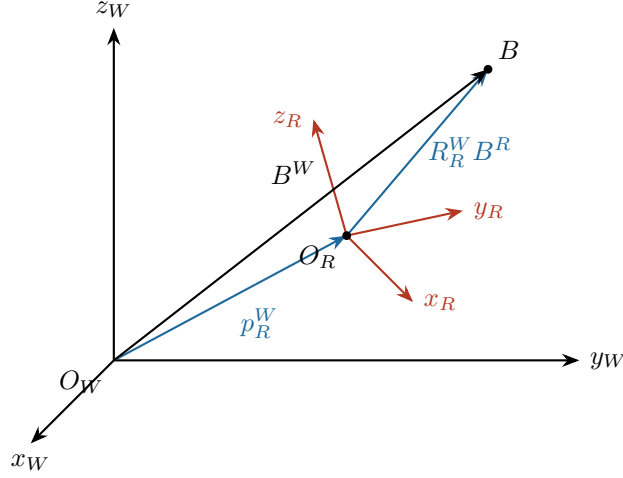


Figure 5.1: Spatial coordinate transformation. The displacement from O_W to B is the sum of the displacement to O_R and the displacement from O_R to B . All three labeled displacement vectors use world components. The perspective sketch illustrates the geometry, not the numerical example.

5.1.1 Worked example

Take

$$p_R^W = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad B^R = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \quad R_R^W = R_z(\pi/4) = \begin{bmatrix} c & -c & 0 \\ c & c & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad c = \frac{\sqrt{2}}{2}.$$

The rotation is about z , chosen to simplify the arithmetic; the transformation law applies equally to a general spatial rotation. Substitution gives

$$B^W = \begin{bmatrix} 2c \\ 2c \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 + \sqrt{2} \\ \sqrt{2} \\ 3 \end{bmatrix}. \quad (5.2)$$

5.2 The reverse coordinate transformation

To recover the coordinates in R , first subtract the origin's world position and then apply the inverse rotation:

$$B^R = (R_R^W)^T (B^W - p_R^W). \quad (5.3)$$

The order matters. The subtraction takes place in W ; the transpose then converts that displacement into R components. For the example,

$$B^R = \begin{bmatrix} c & c & 0 \\ -c & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

5.3 Homogeneous transformations in space

Introduce homogeneous point coordinates by appending a one:

$$\tilde{B}^R = \begin{bmatrix} B^R \\ 1 \end{bmatrix} \in \mathbb{R}^4.$$

The spatial homogeneous transformation is

$$H_R^W = \begin{bmatrix} R_R^W & p_R^W \\ 0_{1 \times 3} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4}, \quad \boxed{\tilde{B}^W = H_R^W \tilde{B}^R.} \quad (5.4)$$

The lower-left block contains three zeros. Block multiplication reproduces Equation (5.1) in the first three rows and preserves the final one. Lines drawn between blocks are visual guides, not extra operations.

For the worked example,

$$H_R^W = \begin{bmatrix} c & -c & 0 & 1 \\ c & c & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad H_R^W \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + \sqrt{2} \\ \sqrt{2} \\ 3 \\ 1 \end{bmatrix}.$$

The last entry is a consistency check for this representation of a point under a rigid transformation. A direction vector is instead augmented with zero; then multiplication rotates the direction without adding a translation.

5.3.1 Composition follows the frame chain

The advantage of homogeneous coordinates remains the same as in the planar case. For an intermediate frame R and a sensor frame S ,

$$H_S^W = H_R^W H_S^R. \quad (5.5)$$

Expanding the blocks gives

$$H_S^W = \begin{bmatrix} R_R^W R_S^R & R_R^W p_S^R + p_R^W \\ 0_{1 \times 3} & 1 \end{bmatrix}.$$

Thus the rotation composes by multiplication, while the sensor-origin displacement must be rotated into W before adding p_R^W . The rightmost map acts first: $S \rightarrow R \rightarrow W$. This is the spatial version of the earlier frame-chain rule.

5.4 Inverting a homogeneous transformation

Rearranging Equation (5.3) separates the reverse rotation and translation:

$$B^R = (R_R^W)^\top B^W - (R_R^W)^\top p_R^W.$$

Consequently,

$$\boxed{H_W^R = (H_R^W)^{-1} = \begin{bmatrix} (R_R^W)^\top & -(R_R^W)^\top p_R^W \\ 0_{1 \times 3} & 1 \end{bmatrix}.} \quad (5.6)$$

This uses the structure of a rigid transformation instead of a general 4×4 inversion. Multiplying the original and inverse blocks gives $R_R^W (R_R^W)^\top = I_3$ and translation $-R_R^W (R_R^W)^\top p_R^W + p_R^W = 0$, verifying the result.

5.4.1 Why reversing the translation requires a rotation

The upper-right block of the inverse is

$$\boxed{p_W^R = -(R_R^W)^\top p_R^W}. \quad (5.7)$$

Geometrically, the displacement from O_R to O_W is opposite to the displacement from O_W to O_R . However, p_R^W uses world components, whereas p_W^R uses robot components. Negating the former reverses the physical displacement but leaves its components expressed in W . The transpose is needed to express them in R .

In the example,

$$p_W^R = - \begin{bmatrix} c & c & 0 \\ -c & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -c \\ c \\ -2 \end{bmatrix},$$

which differs from $-p_R^W = [-1 \ 0 \ -2]^\top$. The full reverse transformation is

$$H_W^R = \begin{bmatrix} c & c & 0 & -c \\ -c & c & 0 & c \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Applying it to $[1 + \sqrt{2} \ \sqrt{2} \ 3 \ 1]^\top$ returns $[2 \ 0 \ 1 \ 1]^\top$, as required.

5.5 What rounding changes

Replacing $c = \sqrt{2}/2$ by 0.7 makes the forward calculation especially simple. Denote the resulting approximate rotation by

$$\hat{R} = \begin{bmatrix} .7 & -.7 & 0 \\ .7 & .7 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

With the same translation and input point, it gives

$$\hat{B}^W = \hat{R} B^R + p_R^W = \begin{bmatrix} 2.4 \\ 1.4 \\ 3 \end{bmatrix}.$$

If we now use the transpose as though it were the inverse, the return calculation produces

$$\hat{R}^\top (\hat{B}^W - p_R^W) = \begin{bmatrix} .7 & .7 & 0 \\ -.7 & .7 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.4 \\ 1.4 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.96 \\ 0 \\ 1 \end{bmatrix}.$$

The discrepancy has a precise explanation:

$$\widehat{R}^\top \widehat{R} = \text{diag}(.98, .98, 1) \neq I_3. \quad (5.8)$$

After rounding, $\widehat{R}$ is not an exact rotation matrix, and its transpose is not its exact inverse. The return calculation scales the first two input components by 0.98. The exact radical form returns the original point without this error.

This deliberately coarse rounding illustrates why orthogonality matters. Using the actual algebraic inverse of $\widehat{R}$ would reverse its own forward map exactly in exact arithmetic, but that map would still fail to be a rigid rotation. In numerical work, keep adequate precision and monitor the rotation constraints rather than assuming that a matrix labeled as a rotation automatically satisfies them.

5.6 Position, orientation, and pose

Position specifies the location of a chosen body-fixed origin relative to a reference frame. In space it has three degrees of freedom. *Orientation* specifies how the body-fixed axes are aligned relative to the reference axes and has three more continuous degrees of freedom. Together, position and orientation define the body's *pose*. An unconstrained rigid body in space therefore has six degrees of freedom.

The number of stored components need not equal the number of degrees of freedom. A rotation matrix contains nine entries subject to six independent orthonormality constraints, with determinant +1 selecting proper rotations. A unit quaternion has four components subject to one norm constraint. Both describe three continuous degrees of freedom. RPY angles use three parameters directly, with the singularity and nonuniqueness discussed in Chapter 4.

Pose representation	Stored entries	Main condition or limitation
H_R^W	16	Rigid-transform block structure
Position and RPY	6	Angle ranges and singularities
Position and quaternion	7	Unit quaternion; q and $-q$ equivalent

For example, two vector representations are

$$\begin{bmatrix} p_R^W \\ \varphi_R^W \\ \theta_R^W \\ \psi_R^W \end{bmatrix} \in \mathbb{R}^6, \quad \begin{bmatrix} p_R^W \\ q_R^W \end{bmatrix} \in \mathbb{R}^7,$$

where q_R^W represents the same orientation as R_R^W under the scalar-first convention. Stacking the components does not make pose composition ordinary vector addition. To combine poses, use the appropriate rotation and translation rules, conveniently encoded by homogeneous matrices.

Once both the position and orientation of a body-fixed frame are specified, the locations of all points fixed to that rigid body follow from their constant coordinates in that frame. This is why a single pose describes the placement of the entire rigid body.

5.7 Additional exercises

Use right-handed frames and exact values unless rounding is explicitly requested.

1. **A spatial point transformation.** Let $R_R^W = R_x(\pi/2)$, $p_R^W = [1 \ -2 \ 3]^\top$, and $B^R = [2 \ 1 \ 0]^\top$. Find B^W and recover B^R using the reverse map.
2. **Homogeneous form.** Build H_R^W for Exercise 1 and use it to transform $\tilde{B}^R$. Build $(H_R^W)^{-1}$ using its blocks and verify that their product is I_4 .
3. **Reversing a displacement.** For $R_R^W = R_z(\pi/2)$ and $p_R^W = [2 \ 1 \ 3]^\top$, compute p_W^R . Compare it with $-p_R^W$ and explain the frame difference. Verify that $R_R^W p_W^R + p_R^W = 0$.
4. **A sensor frame.** Suppose $R_R^W = R_z(\pi/2)$, $p_R^W = [1 \ 0 \ 2]^\top$, $R_S^R = I_3$, and $p_S^R = [1 \ 0 \ 1]^\top$. Find H_S^W and the world coordinates of $B^S = [0 \ 2 \ 0]^\top$. Check your result by transforming first to R and then to W .
5. **Points and directions.** Using the transformation in Exercise 1, transform the points $A^R = [0 \ 0 \ 0]^\top$ and $B^R = [2 \ 1 \ 0]^\top$. Show that $B^W - A^W = R_R^W(B^R - A^R)$. Explain why a homogeneous direction uses a final zero instead of a one.
6. **Rounding and orthogonality.** Replace c in the worked example by $a = 0.707$. Derive $\hat{R}^\top \hat{R} = \text{diag}(2a^2, 2a^2, 1)$. Find the result of the forward map followed by the transpose-based reverse map for $B^R = [2 \ 0 \ 1]^\top$, and compare with the $a = 0.7$ result.
7. **Components versus degrees of freedom.** Explain why position plus a unit quaternion stores seven numbers but describes six continuous degrees of freedom. Normalize $q = [2 \ 0 \ 0 \ 2]^\top$. Use Appendix A to find its rotation matrix, then build a homogeneous transformation with position $[1 \ 0 \ 2]^\top$.

Chapter 6

Robot systems and sensing

Reference frames and transformations describe where a robot and the objects around it are located. A robot must also obtain information about those objects, decide what to do, and produce physical motion. These activities connect the geometric tools of the preceding chapters to the operation of a robot system.

6.1 Sensing, processing, and action

A useful working description of a robot is an intelligent interconnection between sensing and action. Sensors collect measurements; computation uses those measurements to select actions; actuators carry out the actions. The resulting interaction with the environment produces new measurements, closing a feedback loop.

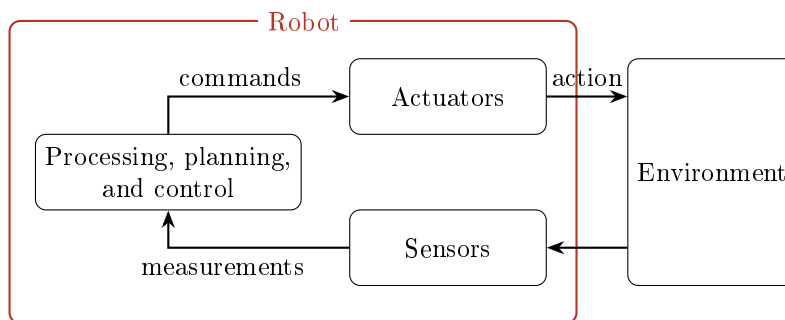


Figure 6.1: A functional view of a robot interacting with its environment. The robot boundary contains sensors, actuators, and the hardware and software that process measurements and generate commands.

6.2 Sensors and the environment

The *environment* includes the objects, surfaces, and people with which the robot interacts. A camera measures incoming light, a microphone responds to pressure variations, and a contact sensor detects

physical contact. Sensors may also measure the robot’s own state; sensing is not limited to observing external objects.

A measurement is not yet an interpretation of the scene. An image records light arriving at the camera; identifying an object and estimating its location require processing that image. Likewise, a contact measurement indicates an interaction at a surface but does not by itself describe the entire surrounding space. Each type of sensor provides particular information that the robot must interpret in relation to its task.

The coordinate transformations developed earlier enter at this point. An object located in a sensor frame may need to be expressed in a robot frame to determine whether it is within reach, or in a world frame to plan motion. The physical object is the same; the coordinate description is chosen to support the calculation being performed.

6.3 Actuators and physical interaction

Actuators produce physical action, for example by driving wheels or moving a manipulator. Wheel-driven motion depends on forces exchanged with the supporting surface. An action may change the environment, the robot’s pose relative to it, or both.

The distinction between sensing and action is a useful functional one: sensors provide measurements, whereas actuators produce physical effects. A motor command can turn a wheel or move a joint, but the resulting motion also depends on the mechanism and its interaction with the surroundings. Moving a mobile robot changes what its sensors observe, even when the objects in the environment remain fixed in the world frame.

6.4 Connecting measurements to commands

The connection between sensing and action includes both computing hardware and software. It may involve interpreting measurements, estimating where objects are, selecting a plan, and producing individual actuator commands. Here, “intelligent” does not require a neural network or machine learning; explicit algorithms and feedback control can also provide this connection.

The central block in Figure 6.1 therefore represents several possible levels of computation. A high-level decision might be to approach an object. More detailed decisions specify how to move, and the commands sent to the actuators must ultimately specify actions that the hardware can execute. All of these functions belong to the connection between measurements and physical action, even when they run in different programs or on different processors.

The enclosing boundary in the diagram is also important: the robot includes the sensors and actuators as well as the computing hardware and software. The central block alone is not the entire robot. The arrows trace a loop through the environment: an action changes the physical situation, sensors measure the resulting situation, and computation can use those measurements to decide the next action.

6.4.1 Example: reaching an object

Suppose a robot must grasp an object that is initially beyond its reach. Sensor measurements first support an estimate of the object's location. The robot must then determine that it needs to move closer, choose an approach, and command its motion before extending the manipulator and closing the gripper. A high-level objective thus becomes a sequence of physical actions, with measurements available to check progress and adjust the commands.

Consider the distinction between the decisions involved. Recognizing the object answers what the robot should interact with. Estimating its position answers where it is. Comparing that position with the robot's reach determines whether the base must move first. Planning selects a sequence of motions, and control turns those motions into actuator commands. These decisions serve the same task but require different information and computations.

After the robot approaches, the object's coordinates relative to the robot have changed. New measurements can update its estimated location before the grasp. This illustrates why sensing and action are connected repeatedly: the robot does not merely observe once and then execute an unrelated motion.

Appendix A

Rotation representation conversions

This reference collects conversions between rotation matrices, roll–pitch–yaw (RPY) angles, and unit quaternions. The matrix and angle conventions agree with Chapter 4, which introduces the practical use of unit quaternions.

Conventions

All frames are right-handed, vectors are columns, and

$$R = R_z(\psi)R_y(\theta)R_x(\varphi) \in SO(3), \quad SO(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = I_3, \det R = 1\}.$$

The angles are roll φ , pitch θ , and yaw ψ , with principal ranges $\varphi, \psi \in (-\pi, \pi]$ and $\theta \in [-\pi/2, \pi/2]$. Quaternions use the scalar-first convention

$$q = [q_w \quad q_x \quad q_y \quad q_z]^T, \quad q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1.$$

The signs in the formulas below specify the quaternion convention; a different component order must be converted before applying them. The quaternions q and $-q$ represent the same rotation. Write $c_a = \cos a$ and $s_a = \sin a$. The function $\text{atan2}(y, x)$ takes its arguments in that order.

RPY angles to a rotation matrix

$$R = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\varphi - s_\psi c_\varphi & c_\psi s_\theta c_\varphi + s_\psi s_\varphi \\ s_\psi c_\theta & s_\psi s_\theta s_\varphi + c_\psi c_\varphi & s_\psi s_\theta c_\varphi - c_\psi s_\varphi \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi \end{bmatrix}.$$

Rotation matrix to RPY angles

For $R = [r_{ij}]$ and $|\theta| < \pi/2$, the angles are recovered by the formulas below. At $\theta = \pm\pi/2$, yaw and roll are not separately determined; a singular-case convention is required. In particular, $\text{atan2}(0, 0)$ has no unique geometric meaning.

$$\begin{aligned} \varphi &= \text{atan2}(r_{32}, r_{33}), \\ \theta &= \text{atan2}\left(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}\right), \\ \psi &= \text{atan2}(r_{21}, r_{11}). \end{aligned}$$

Unit quaternion to a rotation matrix

$$R = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}.$$

This expression assumes a unit quaternion. Normalize a nonzero quaternion before using it; a zero quaternion does not describe an orientation.

RPY angles to a unit quaternion

Using the same intrinsic Z–Y–X convention,

$$\begin{aligned} q_w &= c_{\varphi/2} c_{\theta/2} c_{\psi/2} + s_{\varphi/2} s_{\theta/2} s_{\psi/2}, \\ q_x &= s_{\varphi/2} c_{\theta/2} c_{\psi/2} - c_{\varphi/2} s_{\theta/2} s_{\psi/2}, \\ q_y &= c_{\varphi/2} s_{\theta/2} c_{\psi/2} + s_{\varphi/2} c_{\theta/2} s_{\psi/2}, \\ q_z &= c_{\varphi/2} c_{\theta/2} s_{\psi/2} - s_{\varphi/2} s_{\theta/2} c_{\psi/2}. \end{aligned}$$

Unit quaternion to RPY angles

Away from the pitch singularities,

$$\begin{aligned} \varphi &= \text{atan2}(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)), \\ \theta &= \arcsin(2(q_w q_y - q_z q_x)), \\ \psi &= \text{atan2}(2(q_w q_z + q_x q_y), 1 - 2(q_y^2 + q_z^2)). \end{aligned}$$

For a normalized quaternion, the exact argument of arcsin lies in $[-1, 1]$. In floating-point arithmetic, clamp small roundoff excursions to that interval before evaluating arcsin. This does not resolve the pitch singularity, nor does it replace checking the quaternion's normalization.

Rotation matrix to a unit quaternion

One conversion branch chooses $q_w \geq 0$:

$$\begin{aligned} q_w &= \frac{1}{2}\sqrt{1 + r_{11} + r_{22} + r_{33}}, \\ q_x &= \frac{r_{32} - r_{23}}{4q_w}, \quad q_y = \frac{r_{13} - r_{31}}{4q_w}, \quad q_z = \frac{r_{21} - r_{12}}{4q_w}. \end{aligned}$$

This branch requires $q_w \neq 0$. It fails for an exact half-turn and is numerically sensitive near a half-turn. A robust implementation selects a branch whose denominator is large. The following equivalent identities provide the alternative branches:

$$\begin{aligned} 4q_w^2 &= 1 + r_{11} + r_{22} + r_{33}, & 4q_x^2 &= 1 + r_{11} - r_{22} - r_{33}, \\ 4q_y^2 &= 1 - r_{11} + r_{22} - r_{33}, & 4q_z^2 &= 1 - r_{11} - r_{22} + r_{33}. \end{aligned}$$

Choose a largest squared component, assign its positive square root, and recover the other components from

$$\begin{aligned} 4q_w q_x &= r_{32} - r_{23}, & 4q_x q_y &= r_{12} + r_{21}, \\ 4q_w q_y &= r_{13} - r_{31}, & 4q_x q_z &= r_{13} + r_{31}, \\ 4q_w q_z &= r_{21} - r_{12}, & 4q_y q_z &= r_{23} + r_{32}. \end{aligned}$$

For example, if q_x is chosen, divide the identities for $q_w q_x$, $q_x q_y$, and $q_x q_z$ by $4q_x$ to obtain q_w, q_y, q_z . For a valid rotation at least one component has magnitude at least 1/2. Normalize the resulting quaternion to remove small numerical drift. These formulas assume the input is a proper rotation matrix; they do not repair an arbitrary matrix.